

Daniel Palin

647-993-3596 | danwillpalin@gmail.com | www.linkedin.com/in/daniel-w-palin | danielpalin.github.io

Robotics software engineer working on real-time control, motion planning, and perception for manipulation. Shipped closed-loop visual servoing and sub-millisecond planning for a 6-DOF robotic arm at an NYC robotics startup, and previously led a 14-person ROS 2 navigation team.

EDUCATION

Queen's University

Masters of Engineering in Mechanical and Materials Engineering, GPA: 4.3/4.3

Kingston, Ontario

Sept. 2025 – Aug. 2026

University of Toronto

Bachelor of Arts and Science Major in Physics, Minor in Chemistry

Mississauga, Ontario

Sept. 2020 – May 2025

EXPERIENCE

Robotics Software Engineering Intern

May 2026 – Present

Mbodi AI

New York, NY

- Shipped the company's first real-time closed-loop arm controller: feedforward + PID, visual servoing for a 6-DOF ABB GoFa 10 robotic arm to pick and placed moving objects in manufacturing
- Cut planning latency from 21 ms to ~1 ms per control tick with visibility graphs, RRT-Connect, Pinocchio + DLS
- Implemented joint-space planning to catch whole-arm self-collisions, added rasterized masks for occlusion handling

Technical Lead - Autonomous Navigation Team

Sept. 2025 – August 2026

Queen's Knights Robotics Team

Kingston, ON

- Led 14-person team developing an autonomous robot, coordinating perception, planning, and control integration
- Placed 5th in the world out of 24 teams at Purdue University's Advanced Robotics Competition
- Built ROS2 navigation system fusing camera and LiDAR data with IMU and wheel encoders for localization
- Designed a custom ROS 2 controller plugin for improved mecanum drive navigation

Teaching Assistant - Mechatronics and Robotics Design I

Dec. 2025 – April 2026

Queen's University

Kingston, ON

- Instructed 80 students on systematic debugging of embedded systems, sensor integration, and PID control tuning
- Evaluated technical reports and team projects on control architecture, code quality, and testing methodology

RELEVANT PROJECTS

Vision-Language-Action Model for Manipulation | *Python, PyTorch*

Dec. 2025 – Present

- Built SO-101 robot arm platform and collected 250+ demonstration trajectories for imitation learning
- Fine-tuned SmolVLA vision-language-action model to generate robot actions from visual observations and natural language commands
- Deployed trained model on physical hardware achieving 98% success rate for language-guided pick-and-place tasks
- Posted teleoperation datasets to Hugging Face, receiving 1000+ total downloads

6-DOF Robotic Arm Pick-and-Place using Diffusion | *Isaac Sim, PyTorch, AWS*

Sept. 2025 – Nov. 2025

- Configured and simulated 6-DOF robotic arm in NVIDIA Isaac Sim for manipulation training
- Trained diffusion-based manipulation policies achieving 92% pick-and-place success rate in simulation
- Built GPU-accelerated training pipeline on AWS EC2 processing 6400+ waypoint collisions

Bio-Inspired UV Navigating Robot | *Arduino, UV Sensors*

Sept. 2025 – Dec. 2025

- Developed novel straight-line navigation method using UV sensors with proportional control
- Built physical differential drive test robot and conducted controlled experiments across varying weather and lighting conditions to validate navigation accuracy
- Achieved 48% accuracy improvement (20.4 vs 39.1 inch deviation, $p < 0.05$) over open loop baseline

TECHNICAL SKILLS

Programming: Python, C++, CUDA, Git, Linux

Robotics Simulation: Isaac Lab, Isaac Sim (extensive), Omniverse, Gazebo, MuJoCo, CAD

Robotics Software: ROS2, SLAM, Motion planning, Sensor fusion, GPU-accelerated computing

AI/ML: PyTorch, LeRobot/SmolVLA, Reinforcement Learning (PPO, SAC, MAPPO), Imitation Learning, Diffusion Models, Vision Foundation Models (SigLIP2, DINOv2)